REPORT — Bridge C order 3 (shadow localisation)

Order run as preregistered: scripts hashed (scripts/SHA256SUMS) → Map 1 → sign control → Map 2 → Map 0.

Map 1: decomposition identity (Category A)

Statement. For each piece P = r·e1 + N(21,−1) + N(0,1) with generators u1=(21,−1), u2=(0,1), the order-2 coefficient of entry 107 splits as A1_P = D_P + W_P^{(Q(u2))} + W_P^{(Q(u1))}, where - the drift is D_P = −mean(f)(l1J_a + l2J_b); - the sawtooth terms are W_P^{(Q(u_i))} = −(√π/2)Q(u_i)^{-1/2}·N^{-1}Σ f·B1(z_i/N).

The rays contribute R = Σ_walls ±(√π/2)Q(u)^{-1/2}·mean_u(f).

Proof. In the entry-107 formula, Q(u2) = r and Q(u1) = p. With A = [[2,7],[7,7]]: - Q(0,1) = 7/2; - Q(21,−1) = ½(2·441 − 2·7·21 + 7) = 595/2; - Q(1,0) = 1, which is the generator on the boundary pieces and the walls.

So every W term carries exactly one of the labels Q ∈ {1, 7/2, 595/2}. The decomposition is a regrouping of the finite sum in entry 107, so it is exact. ∎

Numerical check. Over 105 companions × 216 points, the largest gap between the summed parts and order2 A01_float is 1.8e-15. **Pass.**

Sign control (must pass before verdicts are read)

- A0 at w=1: rank 36, joint rank 36 (closed).
- A0 at the wrong sign: joint rank 72 > 36.
- **Control passed.**

Map 2: closure tests (Category B, numerical ranks with SVD threshold 1e-9)

part	rank	joint w=½	joint w=1	prediction	verdict
D	58	116	98	closed at w=1	**FALSIFIED**
R	105	210	210	closed at w=½	**FALSIFIED**
W[1]	90	178	156	not closed	consistent
W[7/2]	60	116	120	not closed	consistent
W[595/2]	100	200	200	not closed	consistent
W total	105	210	210	not closed	consistent
A1	105	210	210	(order-2 result: not closed)	reproduced

Overall: the preregistered claim, that the defect sits only in W while D and R are separately S-closed, is **falsified**. None of the parts closes on its own. The shadow is not located in the wall sawtooth by this split.

Map 0: numeric anchor

Data from the orthant/cone series, fitted on the preregistered grid t_j = 2^{-j}·0.5/(297.5·(2m)²). The fitted t^{-1/2} coefficient matches A1 at all x ∈ {1, 1/2, 1/3, 2/3, 1/4, 1/5, 1/6}: - largest relative error: 7.5e-6 (threshold 1e-3); - **7/7 pass.**

The O(1) coefficient A2 (descriptive only) is in data/map0.json. - x=1: −0.323 − 0.445i - x=1/2: ≈ 0 - x=1/6: 2.33 + 1.89i

Order 3 notes (descriptive)

- **Ranks.** D has rank 58 and R has rank 105, both below the full 216-row space, so non-closure is a real result, not a saturation artefact. The ranks are numerical (Category B), and none of them was certified by mod-p arithmetic.
- **Reading.** The S-transform mixes drift, rays and walls. A split by geometric origin is not modular-covariant, so the shadow has to be looked for in a different basis. A natural next candidate is a decomposition by theta characteristics or Q-classes across all three parts, rather than by piece geometry.
- **Map 0.** Map 0 is an independent check that the t^{-1/2} coefficient, and hence the order-2 formula, is right on the 2m-scaled grid. The post-hoc issue from order 2 did not recur.
- **Still open.** Identification of Ξ, the analytic O(1) term, and smoothness of any defect.
- **Process.** No PREREG change. No pushes from order3.